Sam (Peset) Tan

Senior Software Engineer

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SUMMARY

Senior software engineer building healthcare applications and shared platform infrastructure for medication-management workflows. Hands-on experience with Omnicell's OmniSphere platform, closed-loop dispensing integrations, restocking, diversion, and transactional auditing. Combines C#/.NET and distributed-systems development with technical leadership of three other engineers.

TECHNICAL SKILLS

Backend: C#, .NET, ASP.NET Core, TypeScript, Node.js, Python, Go

Systems and data: Kafka, PostgreSQL, ClickHouse, Redis, Elasticsearch, multitenancy, transactional outbox

Cloud and delivery: AWS, Kubernetes, Helm, Codefresh CI/CD, NiFi

Frontend: Angular, Nx, reusable components

Quality: Unit/integration/acceptance tests, concurrency testing, observability, performance benchmarking

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Omnicell | 2017-Present

Healthcare cloud software supporting medication workflows, dispensing integrations, and shared configuration, identity, and audit services.

- **Cloud modernization.** Served as inaugural developer for a hospital restocking cloud application; helped migrate desktop workflows into three backend services and two Angular applications on Kubernetes. Owned scheduled-restocking design and much of its implementation, refining failure handling through manual and automated tests.
- **Shared audit infrastructure.** Implemented reusable ASP.NET Core request auditing and entity-change capture with tenant/request isolation and preserved package contracts. Optimized the authenticated single-role middleware path to execute 32.6% faster with 28.2% fewer allocations in repeated in-process Release/.NET 10 benchmarks.
- **Reliable event delivery.** Contributed to shared .NET/PostgreSQL/Kafka transactional-outbox infrastructure with retries, leases, and per-key ordering. Optimized backlog queries; synthetic local tests across 100 tenants and three relays reconciled 299,100 events with no observed loss, duplication, or ordering errors, while retaining at-least-once delivery semantics.

Omnicell | Senior Software Engineer (continued)

- **Audit ingestion and compatibility.** Implemented TypeScript/Node.js ingestion from tenant-specific Kafka topics into ClickHouse, with payload validation, deterministic event IDs, and offset commits after persistence. Retained legacy Elasticsearch integration and diagnosed cross-service delivery gaps; guided schema and message compatibility with existing systems.
- **Performance and production reliability.** Added edge-sync instrumentation and changed blocking Kafka producer behavior to reduce message lag. Integrated Redis caching into shared configuration infrastructure and coordinated Elasticsearch reindexing with SRE through production to address capacity and operational-cost concerns.
- **Cross-service delivery.** Refined stock-area transfer designs and coordinated integration and testing across teams through initial release. Contributed to historical-data hydration design and coordination for closed-loop inventory processing, spanning staging, enrichment, and progress tracking.
- **Full-stack engineering.** Built Angular/Nx audit-search behavior and reusable frontend components; preserved selection across pagination and reset it on filter/sort changes. Added role-change and login auditing, readable role names, and privacy metadata across service integration.
- **Technical leadership.** Technical lead for three other engineers since 2025 while continuing hands-on development. Review team-proposed designs, mentor engineers into broader backend ownership, and promote constructive feedback, learning from mistakes, and shared production support. Collaborate with product, architects, customers, and other teams on priorities and technical tradeoffs.

Web Developer and Data Analyst | ATAC | 2008-2017

- Developed C# tools to parse large aviation datasets, prepare Excel outputs, and integrate with proprietary Windows applications for analysts and airspace controllers.
- Cleaned and presented aviation data; prototyped Angular applications with Node.js backends to visualize flight, noise, weather, and historical trends.

EDUCATION

M.S. Software Engineering | Penn State

B.S. Electrical Engineering | San Jose State